

A cautionary tale: Why you shouldn't trust random measurements of the energy/water/etc. use of AI even if they're often repeated.

Posted by u/Tyler_Zoro • 0 points • 37 comments

You've probably heard the claim:

> More like 10 Google searches per chat prompt.

That's a direct quote from another user in this sub talking about the power usage of AI. I asked them to cite their source. They came back with seven links, and zero commentary on why they felt those links backed up their claim.

Okay though, seven sites all made the same claim. It must be true, right?

Well... about that. I looked into the links provided. I won't go into all the long analysis I wrote back to them, but here are the highlights:

- * Some of the links were to pages that repeated the same phrase, but either cited no source or attempted but failed to cite a source (one even linked to a `file:///` link that went to a file on their personal PC's downloads folder!)
- * Some of the links were to horrifically bad articles that clearly had no idea what they were talking about, and which also did not cite their sources.
- * But one article (Washington Post, though paywalled) did cite a source (an "autonomous intergovernmental organisation" called "The International Energy Agency" in their policy advisory paper, "Electricity 2024: Analysis and forecast to 2026").

The IEA said:

> Search tools like Google could see a tenfold increase of their electricity demand in the case of fully implementing AI in it. When comparing the average electricity demand of a typical Google search (0.3 Wh of electricity) to OpenAI's ChatGPT (2.9 Wh per request), and considering 9 billion searches daily, this would require almost 10 TWh of additional electricity in a year.

Well that seems kind of definitive right? You would think. That claim is sourced to a paper by

Alex de Vries which says:

> Alphabet's chairman indicated in February 2023 that interacting with an LLM could "likely cost 10 times more than a standard keyword search." As a standard Google search reportedly uses 0.3 Wh of electricity, this suggests an electricity consumption of approximately 3 Wh per LLM interaction.

I'm going to ignore the second half of that which is a guess at how much power Google uses for search, and is very, very rough. But the first part cites a comment from a Google executive. In [this Reuter's article](<https://www.reuters.com/technology/tech-giants-ai-like-bing-bard-poses-billion-dollar-search-problem-2023-02-22/>) the following is said:

> In an interview, Alphabet's Chairman John Hennessy told Reuters that having an exchange with AI known as a large language model likely cost 10 times more than a standard keyword search, though fine-tuning will help reduce the expense quickly.

Finally, we find the source of the 10x claim. In February of 2023, an off-hand comment indicated that Google's raw costs would go up 10x if they naively implemented ChatGPT-3 or its equivalent into search results (not Google's AI summary which is a cut-down model merely summarizing existing results pages).

There's no sense given of what constitutes the "exchange" he's talking about and if it's just one prompt, or how much those costs relate to power vs. hardware changes vs. staffing vs. lack of edge-hosting for AI, etc.

It's just a very high-level cost comparison that, since it is impossible to verify, is at best highly questionable.

Yet it's repeated as fact that this means that any request to an AI is using 10x the electricity of a Google search, over and over and over again by people who don't know any better.

TL;DR: No AI company publishes their technical details on how much it costs to run their infrastructure, so everyone who makes a claim that relies on that info is, at best, guessing. They might be citing some cargo-culted guess that someone else made a long time ago, but ultimately it's just a guess.

Comments

u/GeorgeRRHodor • 0 points • 2025-12-03 11:28 UTC

Well, let's assume it's just 3x, then; not 10x. And given that any AI result in search needs a keyword search to get started, that's probably a wide undershot.

If search traffic meaningfully includes AI result, a three (or even two) times increase in energy consumption is still an issue.

u/Tyler_Zoro • 1 points • 2025-12-03 13:28 UTC

> If search traffic meaningfully includes AI result, a three (or even two) times increase in energy consumption is still an issue.

Why? Here's a thought: how much power does visiting each of the top 5 pages, with their massive real-time auctions for placing ads cost? If even 10% of users don't click on those results and get their answer from the summary, it's probably a wash, if not a win.

u/TheBrightMage • 2 points • 2025-12-03 09:11 UTC

>That's a direct quote from another user in this sub talking about the power usage of AI. I asked them to cite their source. They came back with seven links, and zero commentary on why they felt those links backed up their claim.

Well, in theory, internet is supposed to make human knowledgable by expanding their access to knowledge. Practically, however, it gives the untrained mass access to more GARBAGE, of which these mass have no relevant training to screen out. This applies to any and all things that requires objective data in general, and even experts can fall into the trap of personal bias. "Citing Sources" especially scientific ones are skills that requires proper training.

Life Cycle Impact assessment requires expert knowledge and skills (and correspondingly, high cost), which paywalls the proper of analysis of data from Laymen. It's also barely possible to explain statistical analysis properly to someone without training in statistics

u/Tyler_Zoro • 1 points • 2025-12-03 13:32 UTC

I agree that people should have better critical thinking skills. I don't think that has anything to do with the internet or AI though.

u/TheBrightMage • 1 points • 2025-12-03 14:01 UTC

The root cause is human nature, yes. But these are amplifier. Call me elitist, but when information access was limited, misinformation spread was in control. Nowadays, it just leaks out. In the case you presented of 7 irrelevant citations, I'm guessing that these sources comes from "Journalist" and are not peer-reviewed, which is sadly, too common. Internet and AI, as much as it can amplify the capability of experts, also amplifies the ability of any laymen to produce misinformation as well.

u/MegaMangus • -1 points • 2025-12-03 08:29 UTC

Even without actual data, it just feels weird that ProAI feels so comfortable throwing around that AI cost is relatively nothing. Like I guess it isn't impossible but the fact that running LLMs locally is not within the reach of mid specs computers and the fact that most if not all the data used during the conversation is stored to use for further training feels to me as pretty clear indicators that the energy usage is not negable, and if that's the case there needs to be a way to cool the system using that energy.

u/Gimli • 1 points • 2025-12-03 15:01 UTC

Running LLMs locally isn't limited by energy but by VRAM. You need quite a lot of it. Smaller models can be run on consumer hardware, if you want something good that's where it gets

trickier. But even there the costs aren't that significant. You'd have a hard time making ChatGPT burn more power than you do by gaming, because most people don't just sit there plugging stuff into ChatGPT all day. They make a few requests then do something with the results.

u/Tyler_Zoro • 3 points • 2025-12-03 13:34 UTC

> Even without actual data, it just feels weird that ProAI feels so comfortable throwing around that AI cost is relatively nothing.

The claim is that AI energy costs are an existential threat. That claim requires evidence. The evidence turns out to be, "we didn't have data, so we guessed."

That's the issue here. The burden of proof is on the person making the claim.

u/MegaMangus • 0 points • 2025-12-03 14:46 UTC

No necessarily against the first point but I want to point out something about the second.

You are confusing the requirements for burden of proof in criminal cases with administrative handling. In any kind of administrative handling from any responsible society you have to proof beforehand that any project you try to implement is done with a sensible managing of the resources at hand for it to be approved. It then can and should be asked to provide with updated data at any point so the public have the knowledge if the accepted budget (although it is not necessarily talking only about economy so I am not sure if that's the word) is being followed through in reasonable reach from the original statement that was accepted.

I get that the melodramatic tone of the whole discussion reminds of someone demanding for proof of not being guilty, but is a completely normal procedure that has nothing to do with prosecution and more in line with responsible project handling.

u/Tyler_Zoro • 3 points • 2025-12-03 15:46 UTC

> You are confusing the requirements for burden of proof in criminal cases

I was not, at any time, referring to criminal law. I was speaking only about rational discourse.

There was a claim made. There was no evidence for that claim. That's the whole thing, done.

> In any kind of administrative handling from any responsible society you have to proof beforehand that any project you try to implement is done with a sensible managing of the resources at hand for it to be approved.

That's as may be, but you can't just say, "the Empire State building wasn't built with the right kind of stone!" because that's not evidence. Sure, it's absolutely on the engineers to have chosen the right kind of stone for the facade, but you can't just make the claim that they didn't do that correctly and then demand that they prove you wrong. Why are you

claiming that? Do you have evidence for your claim?

Justify your claim. That's all anyone is asking.

u/MegaMangus • 2 points • 2025-12-03 16:32 UTC

Actually, not sure if for the times of the empire state building was true, but buildings is a good analogy for my point. Construction isn't just improvised, it is project with a lot of measurements in terms of scale, cost in personal, cost in material, etc. If at any point before, during or after the construction there is a reasonable doubt that the approved project isn't being followed the people pushing the project do have the duty to prove the project was or is being followed as initially planned within reasonable distance (and with the difference in distance properly justified).

So yeah, you can say "the empire state building wasn't made with the right stone" to which the answer is "actually here is the data of why we didn't used that" and answering "well tough luck lol" is considered a proof of negligence.

So if you are not using it in the criminal sense you may want to choose a different expression, as the criminal law angle of saying "burden of proof" is unavoidable while not holding the same weight in this discussion.

u/ifandbut • 3 points • 2025-12-03 11:10 UTC

>AI cost is relatively nothing.

It cost me less than a cent to make an image on my local computers with FOSS AI.

>running LLMs locally is not within the reach of mid specs computers

It very much is. My potato of a work laptop with a Nvidia Quadro or something bottom of the barrel from 5+ years ago runs it fine. Might take a minute or two to make an image, but it runs and was simple to setup and install. The longest part of the setup was just downloading the weight files.

>most if not all the data used during the conversation is stored to use for further training

Gonna need a source for that. But why would saving a text file take any more power than sending an email?